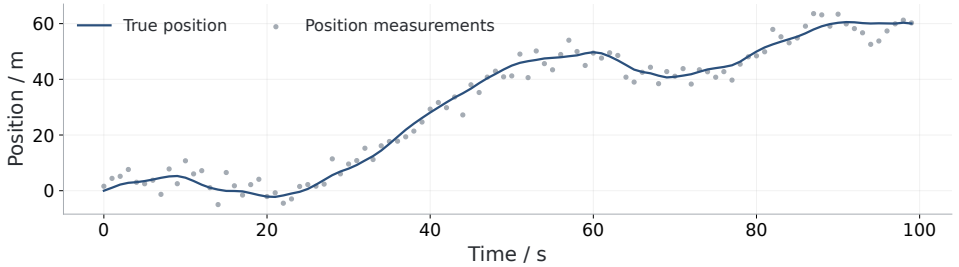
How Kalman Filters Track Noisy Motion

Prediction, uncertainty, and measurement updates

SKILL Application Test

A position sensor, a moving object, and a reusable estimator.

Track the motion behind noisy measurements



The observable signal

100 samples at 1-second intervals. The sensor measures position with standard deviation $3\,\mathrm{m}$.

The estimation task

Recover position and velocity, quantify their uncertainty, and continue through missing observations.

Represent the state and the measurement separately

State: what the system is doing

$$x_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}, \qquad \hat{x}_k = \begin{bmatrix} \hat{p}_k \\ \hat{v}_k \end{bmatrix}.$$

p_k: position in m.
v_k: velocity in m/s.
A hat marks an estimate. The subscript k identifies the sample time.

Measurement: what the sensor sees

$$z_k = Hx_k + e_k, \qquad H = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

The sensor observes position; velocity enters through the motion model over time.

$$e_k \sim \mathcal{N}(0, R), \qquad R = 9 \text{ m}^2.$$

Measurement variance is the square of its standard deviation.



A motion model carries information into the next step

Predict motion, allow acceleration

$$x_k = Fx_{k-1} + Ga_{k-1},$$

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix}.$$

Random acceleration a_{k-1} changes both position and velocity over the interval.

Give model uncertainty a scale

$$a_{k-1} \sim \mathcal{N}(0, q), \qquad Q = qGG^T.$$

At $\Delta t = 1$ s and $q = 0.2\ (\mathrm{m/s^2})^2$,

$$Q = \begin{bmatrix} 0.05 & 0.10 \\ 0.10 & 0.20 \end{bmatrix}.$$

The tracking experiment uses independent acceleration and sensor noise across time.

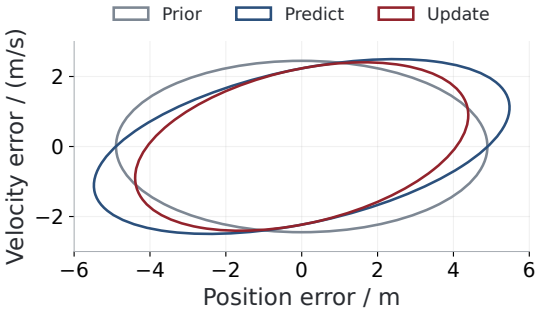
The units follow the state components

Q_{11} has units m^2, Q_{22} has units $(\mathrm{m/s})^2$, and the cross terms have units m^2/s.

Covariance describes the shape of uncertainty

$$P = \mathbb{E}[(x - \hat{x})(x - \hat{x})^T] = \begin{bmatrix} \sigma_p^2 & \sigma_{pv} \\ \sigma_{pv} & \sigma_v^2 \end{bmatrix}.$$

The diagonal entries describe component variances. The off-diagonal term links position error to velocity error.
A tilted ellipse means that a position correction also contains information about velocity.



This geometry starts from $P = \mathrm{diag}(4, 1)$, predicts for 1 s with $q = 0.04$, and updates with $R = 9$.

Contours enclose 95% of the joint Gaussian probability: $(x - \hat{x})^T P^{-1} (x - \hat{x}) = 5.991$.

First combine two scalar estimates

A fully worked position update

Superscripts $-$ and $+$ mean before and after the measurement update. Prior: $\hat{p}^- = 10$ m, $P^- = 4$ m^2.

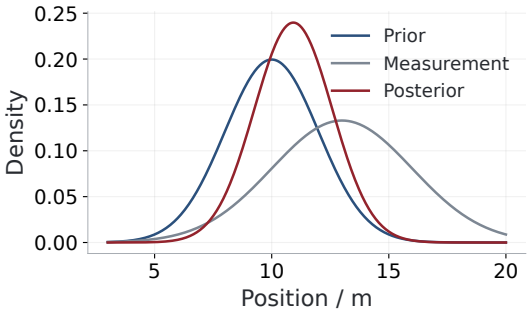
Measurement: $z = 13$ m, $R = 9$ m^2.

$$K = \frac{P^-}{P^- + R} = \frac{4}{13} \approx 0.308.$$

$$\hat{p}^+ = 10 + \tfrac{4}{13}(13 - 10) \approx 10.923 \text{ m}.$$

$$P^+ = (1 - K)P^- \approx 2.769 \text{ m}^2.$$

The estimate moves 0.923 m toward the observation.



The posterior combines the prior density and the measurement likelihood.

Predict both the state and its covariance

Propagation equations

$$\hat{x}_k^- = F\hat{x}_{k-1}^+, \qquad P_k^- = FP_{k-1}^+F^T + Q.$$

The superscript $-$ means before the current measurement; $+$ means after the update.
Prediction transports the previous uncertainty and adds process uncertainty.
In the position component,

$$P_{11}^- = P_{11}^+ + 2\Delta t P_{12}^+ + \Delta t^2 P_{22}^+ + Q_{11}.$$

One numerical step

Take $\Delta t = 1$ s and, for this calculation, $q = 0.04$.

$$\hat{x}^+ = \begin{bmatrix} 10 \\ 2 \end{bmatrix}, \quad P^+ = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\hat{x}^- = \begin{bmatrix} 12 \\ 2 \end{bmatrix}, \quad P^- = \begin{bmatrix} 5.01 & 1.02 \\ 1.02 & 1.04 \end{bmatrix}.$$

Position and velocity become correlated because uncertain velocity affects the next position.

The innovation measures a prediction error

Measure the discrepancy

$$y_k = z_k - H\hat{x}_k^-, \qquad S_k = HP_k^-H^T + R.$$

y_k is the innovation. S_k includes uncertainty in the prediction and in the sensor.
The standardized innovation $y_k/\sqrt{S_k}$ puts errors on a comparable scale.

Continue the worked example

For $z = 13$ m, $H\hat{x}^- = 12$ m and $R = 9$ m^2,

$$y = 1 \text{ m}, \qquad S = 5.01 + 9 = 14.01 \text{ m}^2.$$

$$y/\sqrt{S} \approx 0.267.$$

A 1 m discrepancy is small relative to the combined predicted uncertainty.

The innovation standard deviation is $\sqrt{S} \approx 3.743$ m. Under the model, $y \sim \mathcal{N}(0, S)$; the 95% reference interval is ± 7.34 m.

One position observation can correct velocity too

The vector Kalman gain

$$K = P^{-}H^{T}S^{-1} = \frac{1}{14.01}\begin{bmatrix}5.01\\1.02\end{bmatrix} \approx \begin{bmatrix}0.3576\\0.0728\end{bmatrix}.$$

The first component converts position innovation into a position correction; the second converts it into a velocity correction.

Correct the prediction

$$\hat{x}^{+} = \hat{x}^{-} + Ky \approx \begin{bmatrix}12.3576\ \text{m}\\2.0728\ \text{m/s}\end{bmatrix}.$$

The positive cross-covariance makes an unexpectedly large position suggest a larger velocity.
K_p is dimensionless; K_v has units s^{-1}.

Same prediction, different sensor variance	R (m^2)	K_p	K_v (s^{-1})
Worked example	9	0.358	0.073
More precise sensor	1	0.834	0.170

Update uncertainty with the corrected state

Joseph form keeps the noise contributions explicit

$$A = I - KH, \qquad P^{+} = AP^{-}A^{T} + KRK^{T}.$$

Both terms are covariance contributions. This form is useful for preserving symmetry and positive-semidefinite structure in numerical implementations.

Before the observation

$$P^{-} = \begin{bmatrix} 5.01 & 1.02 \\ 1.02 & 1.04 \end{bmatrix}.$$

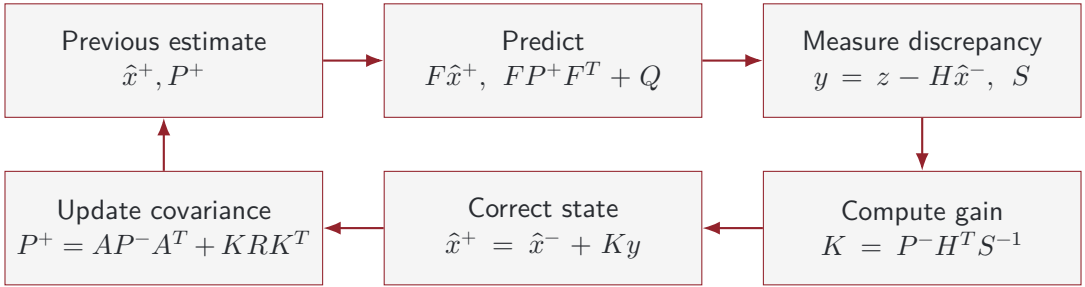
Position SD: $\sqrt{5.01} \approx 2.238\,\mathrm{m}$.

After the observation

$$P^{+} \approx \begin{bmatrix} 3.2184 & 0.6552 \\ 0.6552 & 0.9657 \end{bmatrix}.$$

Position SD: $\sqrt{3.2184} \approx 1.794\,\mathrm{m}$.

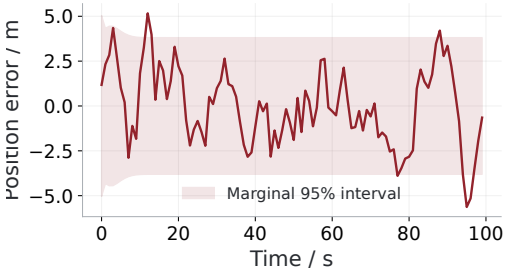
The estimator is a repeating predict-update loop

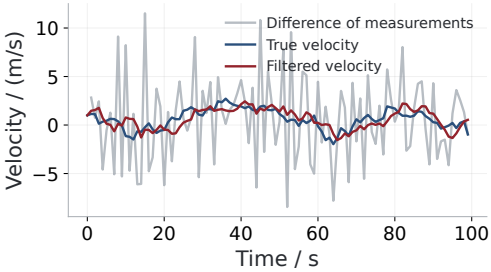


Each cycle carries forward a state vector and a covariance matrix. The innovation connects incoming data to the predicted state.

At a missing observation, retain the prediction and advance to the next timestamp.

The tracking run reduces position error





Position RMSE over steps 10–99

RMSE is the root mean square error.

Measurements: 3.133 m.

Filtered position: 2.143 m.

Read the uncertainty interval

Error means estimate minus truth. The shaded band is $\pm 1.96\sqrt{P_{11}}$; its width comes from the covariance recursion.

Velocity is inferred through motion over time

Direct differences amplify sensor noise

$$\tilde{v}_k = (z_k - z_{k-1})/\Delta t.$$

With independent sensor noise, its noise variance is $2R/\Delta t^2 = 18\ (\mathrm{m/s})^2$.
The Kalman state combines this temporal information with the motion model.

The same run, steps 10–99

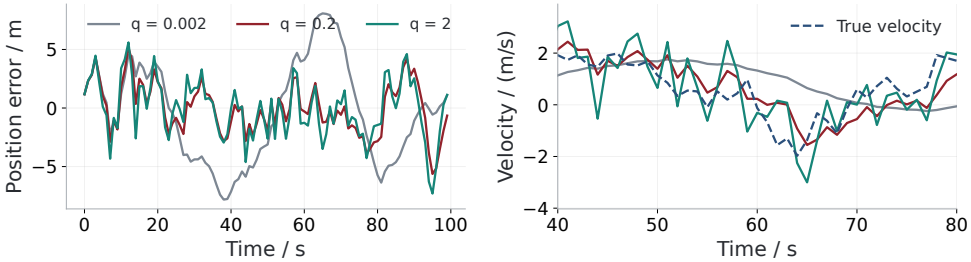
Velocity estimate	RMSE (m/s)
Measurement difference	4.204
Kalman state	0.848

Repeated positions constrain velocity because $HF = [1\ \Delta t]$ includes the velocity component.

Check that the hidden state can affect the observations

For $\Delta t \neq 0$, $\begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & \Delta t \end{bmatrix}$ has rank 2: both state components are observable over time.

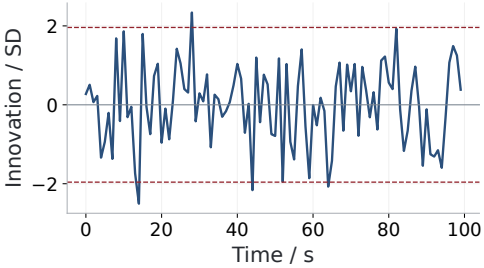
Process noise controls responsiveness

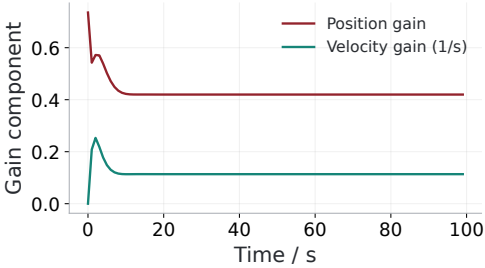


Assumed acceleration variance q	0.002	0.2	2.0
Position RMSE (m)	4.401	2.143	2.445
Velocity RMSE (m/s)	1.169	0.848	1.201

Small q makes the trajectory more persistent; larger q allows faster changes. Here the generating process uses $q = 0.2$.

Read the innovation and gain as diagnostics





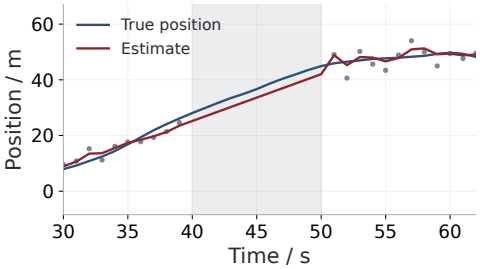
Innovation checks the prediction scale

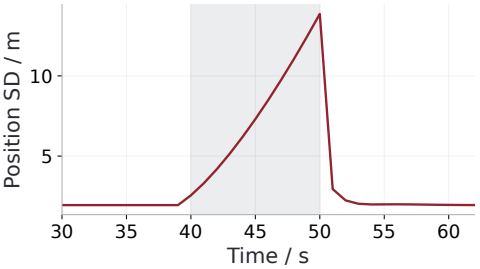
Read $y/\sqrt{S}$ against the dashed ± 1.96 reference. Persistent shifts can reveal a model or sensor change.

Gain records the evolving balance

The position and velocity gains settle as the covariance recursion approaches a stable regime.

Missing measurements increase uncertainty





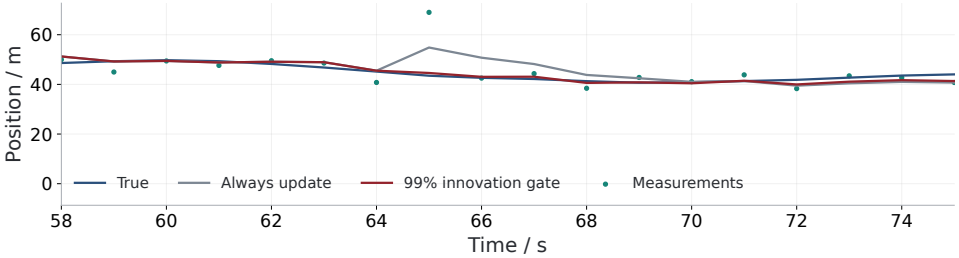
Prediction continues through the gap

Observations at steps 40–50 are withheld.
Apply the prediction equations at every step.

Position standard deviation

Step 39: 1.944 m. Step 50: 13.863 m.
The next observation reduces it to 2.944 m.

Use the innovation scale to screen a large outlier



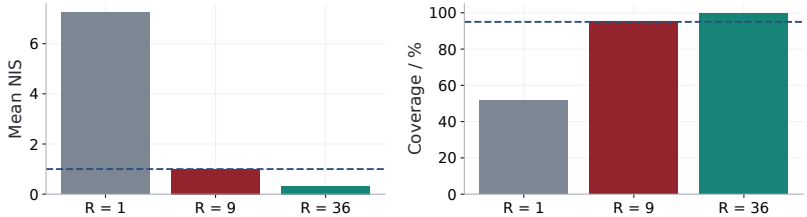
A normalized test

Normalized innovation squared: $\mathrm{NIS} = y^2/S$. For one Gaussian measurement, the 99% χ^2_1 threshold is 6.635.

An injected 30 m error at step 65

Its NIS is 38.42. The gate retains the prediction for this step; later observations resume correction.

Check covariance calibration across repeated trials



400 independent trials, 90 evaluated steps each; generating $q = 0.2$, $R = 9$.

Assumed R	1	9	36
Mean NIS	7.232	1.007	0.323
95% position-interval coverage	51.7%	95.1%	99.8%

Coverage is the fraction of true positions inside the estimated intervals. The matched model reaches 95.1%.

Translate the equations into a reusable update

```
def step(x, P, z, F, Q, H, R):
    xp = F @ x
    Pp = F @ P @ F.T + Q
    if z is None:
        return xp, Pp
    y = z - H @ xp
    S = H @ Pp @ H.T + R
    K = np.linalg.solve(
        S, H @ Pp).T
    x = xp + K @ y
    A = np.eye(len(x)) - K @ H
    P = A @ Pp @ A.T + K @ R @ K.T
    return x, (P + P.T) / 2
```

A clear array contract

x: length 2; P, F, Q: 2×2.
z: length 1; H: 1×2; R: 1×1.
Solve a linear system for the gain. Use Joseph form for the covariance, then remove numerical asymmetry.
The caller supplies elapsed-time-specific F and Q and handles any gating policy.
Earlier measurements are summarized by x and P; estimator storage stays fixed at the state size.

Build the estimator around a testable motion model

Design decision	What to establish
State and sensor	Name the physical variables, units and sensor mapping H.
Timing	Build $F(\Delta t)$ and $Q(\Delta t)$ from the motion assumptions.
Uncertainty	Set P_0, estimate R, and tune Q using errors and innovations.
Data handling	Continue prediction through gaps; define explicit outlier handling.
Validation	Compare estimates with truth and test interval coverage across trials.

The filter carries two things forward: an estimate of the state and a quantitative account of its uncertainty.

Further reading: Welch and Bishop, *An Introduction to the Kalman Filter*; MathWorks, Linear Kalman Filters.